**Justification Document – Group 20**

A non-significant result means we detected no effect, not that none exists. Our confidence intervals limit how large a PM2.5 effect can be, but several paths could not be measured with city-wide daily data.

**Why "non-significant" variables might still matter**

| Variable | Our result | What the result cannot rule out |
|---|---|---|
| City-wide PM2.5 effect | Double Machine Learning (DML) [1]: −0.32% per +10 μg/m³ (95% CI −1.83 to +1.21) | A small effect of up to about ±2%, roughly ±2,200 rentals on an average day |
| Han River share on days off | −0.90% (95% CI −2.70 to +0.93); only 18 Bad days off | A day-off effect as large as the workday one (−1.17%) or larger; the interval is too wide to tell |
| Threshold above 35 μg/m³ | Extra slope p = 0.69 | A reaction at very high levels; only 13 days exceeded 50 μg/m³ |
| Previous-day PM2.5 | Week 4 DML: −0.6% to −0.8%, p ≥ 0.36 | A delayed reaction; same-day and previous-day PM2.5 move together (r = 0.65), which blurs the two |
| Alerts and perceived air quality | Not tested (7 PM10 alert days) | Riders may react to warnings rather than to measured PM2.5 |
| Commute hours | Not observable in daily totals | A short drop at peak hours that is offset later in the day |

**Four reasons to keep these variables in view**

1. **Absence of evidence is not evidence of absence.** A null result sets an upper bound on the effect; it does not set the effect to zero. We report "the 95% CI lies within about ±2%" rather than "no effect".
2. **Confounding can hide or create effects.** The Bad-day estimate moved from −10.5% (raw regression) to +19.4% (same-month regression) to +3.2% (weather and calendar regression); DML gives +1.1% (95% CI −4.2 to +6.8). One multiple regression even turned the PM2.5 effect positive and significant (+1.6%, p = 0.027) when humidity was included, and negative without it. A variable whose sign depends on one control is not settled either way.
3. **Daily totals can hide changes in how people ride.** Descriptive trip data (no controls) show slightly shorter trips on Bad days: on workdays 19.5 → 19.1 minutes

and 2.18 → 2.12 km per ride, and on days off 24.7 → 24.3 minutes. People may shorten or reschedule rides rather than skip them, which a daily count cannot detect.

4. **A small relative shift can matter locally.** The workday Han River share fell 1.17% per +10 μg/m³ ($p < 0.001$, below the Bonferroni threshold of 0.003 for our 16 core tests). It was not pre-specified, and absolute Han River rentals showed no detectable change (−0.8%, $p = 0.43$), so it is a lead for future work, not a finding. It suggests that sub-groups of riders may respond even when the city total does not.

**Policy implications if the hidden features are ignored**

These are implications of observational findings, not proven effects. By hidden features we mean the season and weather confounder, unmeasured alerts, and within-day behaviour.

| Who | If hidden features are ignored | Better practice |
|---|---|---|
| Seoul Metropolitan Government (transport and environment) | The raw −10.5% gap is read as “fine dust keeps people off bikes”, and measures target dust days when the real driver is the cold season | Evaluate cycling policy with weather- and season-adjusted estimates; focus winter demand measures on temperature and rain |
| Ttareungyi operator | Bikes are redistributed on PM2.5 forecasts, although our DML estimates put the city-wide effect within about ±2% per +10 μg/m³ | Plan supply on temperature, rain and the calendar (about 90% of the importance in Map C); watch Han River stations on high-PM2.5 workdays |
| Air-quality agencies | Alerts are assumed to change travel behaviour, which no one has measured | Record alert timing so it can be linked to hourly rentals |
| Researchers and data providers | Daily, city-wide totals keep within-day and substitution effects hidden | Release hourly and trip-level data and transit-card links; report confidence intervals and multiple-testing corrections, not only p-values |

If fine dust did push some riders into cars, the cost would be real: losing about 5,200 bike-km/day (our upper-bound estimate per +10 μg/m³) with 30% replaced by car trips would add about 330 $kgCO_2eq$ per day, using the passenger-car factor of 0.2111 $kgCO_2eq$/passenger-km [2]. This is a conditional scenario, not a measurement. Our data show no detectable ride loss.

**Ethics and data governance**

Three ethical points follow from this work. First, rider exposure: a null effect on rentals means riders keep cycling on high-PM2.5 days, so the operator and the city should pair demand planning with exposure information (air-quality alerts at stations, mask guidance) rather than read stable demand as "no harm". Second, privacy: our recommendation to link hourly trip and transit-card data must use aggregated or de-identified records, because individual travel histories can be re-identified. Third, equity: supply decisions based on

city-wide averages may under-serve districts or rider groups who bear more exposure; future work should check results by district.

**Scope of these claims**

All figures come from our 727-day analysis (2024–2025) of public rental, station, air-quality, weather and map data [3]–[8], and from the results table in our GitHub repository (https://github.com/siwoo9573/ACDT-Group20-PM25-Ttareungyi). The trip-length figures are descriptive. The data are observational and many models were run, so the exploratory results are hypotheses to test, not conclusions. All estimates are adjusted associations; a causal reading assumes no unmeasured confounding. Each number is listed with its source script and specification in Group20_results_table.xlsx (sheet "Report_numbers"). Significance level: $\alpha = 0.05$ (one-sided for the pre-specified test), with a Bonferroni threshold of $0.05/16 \approx 0.003$ for the 16 core tests. Prediction models were validated by 5-fold cross-fitting (out-of-fold $R^2 = 0.85$ for rentals).

# References


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